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Benchmark-Shift-Aware Intrusion Detection for Evolving Network Traffic: Cross-Dataset Generalization, Calibrated Alerting, and Score-Orientation Diagnostics

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Abstract

Modern intrusion detection systems (IDSs) are often evaluated under matched training and test conditions, whereas deployment environments involve changing traffic distributions, heterogeneous feature-generation pipelines, and shifting attack prevalence. This study investigates benchmark-shift-aware intrusion detection through harmonized cross-dataset evaluation of HIKARI-2021, CICIDS2017, and a CICIoT2023 sample subset. Two payload-free feature spaces are constructed: Rich-64 for detailed HIKARI-2021/CICIDS2017 analysis and Minimal-13 for three-way comparison. Using XGBoost, a supervised Transformer, and a masked-feature self-supervised Transformer, we evaluate discrimination, calibration, threshold transfer, alert-budget behavior, chronological robustness, and score-orientation stability. Across five in-domain XGBoost settings, observed false-positive rates were 4.94–5.40%, and F1-scores ranged from 0.507 to 0.995. Under strict Rich-64 HIKARI-2021-to-CICIDS2017 transfer, all models had zero recall at source-derived thresholds, with two showing inverted score orientation. In the reverse direction, XGBoost reached an 18.9% target false-positive rate, while a nominal 5% target-side alert budget yielded F1 = 0.112. Chronological evaluation further showed that improved ranking metrics did not guarantee stable validation-derived operating behavior. The study provides a reproducible diagnostic framework for evaluating IDS robustness under evolving benchmark conditions.

Keywords: intrusion detection system; benchmark shift; cross-dataset generalization; calibrated alerting; score orientation; alert budget; evolving network traffic; domain shift

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1. Introduction

Modern intrusion detection systems (IDSs) operate in networks where traffic composition, applications, protocols, and attack prevalence change over time. Performance measured under matched training and test conditions may therefore overstate deployment robustness: a detector can perform well within one benchmark yet fail on traffic from a different collection scenario or attack mixture, or on features produced by a different pipeline [1–5].

Deployment also depends on more than threshold-free discrimination. Security analysts require calibrated risk estimates, stable thresholds, and predictable alert volumes. After a benchmark shift, a source-derived threshold may suppress attacks or overwhelm analysts with false alarms. Cross-dataset evaluation should therefore examine precision–recall

area under the curve (PR-AUC), receiver operating characteristic area under the curve (ROC-AUC), calibration, threshold transfer, alert-budget behavior, and score-orientation stability.

Recent research has advanced classical, deep, Transformer-based, self-supervised, and domain-shift-aware IDS methods. Yet these approaches are often assessed under matched or weakly shifted protocols. This study instead compares XGBoost, a supervised Transformer, and a masked-feature self-supervised Transformer within a diagnostic framework using harmonized Rich-64 and Minimal-13 payload-free feature spaces across HIKARI-2021, CICIDS2017, and a CICIoT2023 sample subset. It asks how benchmark shift changes cross-dataset generalization, calibrated alerting, threshold transfer, analyst-limited alerting, and score orientation.

1.1. Related Work and Research Gap

Relevant work spans public IDS benchmarks, cross-dataset generalization, calibration and thresholding, deep or self-supervised modeling, and deployment-oriented diagnostics. Their intersection defines the gap addressed here.

Public benchmarks such as HIKARI-2021, CICIDS2017, and CICIoT2023 support reproducible evaluation across enterprise, encrypted, and IoT traffic. However, their collection scenarios, prevalence, labels, and feature pipelines differ, potentially inducing benchmark-specific shortcuts. These differences motivate harmonized payload-free representations and explicit transfer tests [6–10].

Cross-dataset studies test whether models remain robust when the network environment, time period, or traffic-generation regime changes. Because PR-AUC is prevalence dependent, threshold-free scores alone can obscure weak transfer. This study, therefore, reports average-precision lift (AP-lift), defined as PR-AUC minus target attack prevalence, together with ROC-AUC and operating-point metrics [9,10].

Calibration and threshold transfer connect rankings to deployable alerts. A model can rank samples adequately yet fail if its source-fitted probability mapping or 5% false-positive-rate threshold changes meaning in the target domain. We therefore treat sigmoid calibration, transferred thresholds, and workload-aware alerting as primary outcomes.

Deep, Transformer-based, and self-supervised IDS methods can learn complex representations from labeled or unlabeled traffic. Their robustness must nevertheless be tested directly: reconstruction-oriented pretraining may improve representation learning without stabilizing target-domain orientation or threshold transfer [1,2,11–13].

Deployment-oriented diagnostics address analyst workload and score semantics. Label-free target-score quantiles can test whether the highest-ranked events contain a useful concentration of attacks, whereas oracle score inversion can reveal a reversed target-domain signal. The latter requires target labels and is, therefore, diagnostic rather than deployable [5].

Existing studies commonly examine cross-dataset discrimination, probability calibration, alert prioritization, or representation learning as separate problems. This study does not propose a new detector architecture. Its distinctive contribution is to integrate dual-tier payload-free harmonization, matched and strict source-only transfer, source-fitted calibration and 5% FPR threshold transfer, label-free target-score quantile budgets, prevalence-adjusted AP-lift, and explicit separation of operational scores from oracle-only orientation inversion within one reproducible protocol. This integration distinguishes failures in ranking, score semantics, threshold transfer, and analyst-workload management that PR-AUC and ROC-AUC alone do not expose. Table 1 summarizes the comparison.

Table 1. Summary of related work strands and the research gap addressed in this study.

Research Strand	Typical Focus in Prior IDS Studies	Remaining Limitation Under Benchmark Shift	How This Study Addresses the Gap
Public IDS benchmarks	Dataset construction, benchmark description, attack categories, feature extraction, and in-domain evaluation	Benchmark-specific traffic generation, label distribution, and feature pipelines make direct transfer unreliable	Uses HIKARI-2021, CICIDS2017, and the CICIoT2023 sample subset under harmonized Rich-64 and Minimal-13 payload-free schemas
Cross-dataset generalization	Training on one dataset and testing on another, often using threshold-free discrimination metrics	PR-AUC or ROC-AUC alone may hide prevalence effects, threshold failure, and score-orientation instability	Reports PR-AUC, ROC-AUC, AP-lift relative to target prevalence, source-threshold transfer, and orientation diagnostics
Calibration and thresholding	Probability calibration, reliability diagrams, Brier score, ECE, or fixed operating points	In-domain calibration does not guarantee that source-derived thresholds remain valid in target domains	Fits sigmoid calibration on source validation data and evaluates transferred 5% FPR thresholds on target benchmarks
Deep and self-supervised IDS	CNN, RNN, Transformer, contrastive or reconstruction-based representation learning	Advanced models are often assumed to improve robustness without strict benchmark-shift validation	Compares XGBoost, supervised Transformer, and masked-feature SSL Transformer under the same benchmark-shift protocols
Operational alert diagnostics	Alert prioritization, false-positive reduction, and analyst workload	Standard metrics do not directly show how many useful alerts remain under inspection budgets	Evaluates nominal target-side 1%, 5%, and 10% alert budgets using target-score quantile thresholds without target labels for cutoff selection
Score-orientation analysis	Usually implicit or unreported in IDS evaluation	A source-trained detector may assign higher scores to the wrong target-domain class	Uses oracle-flipped diagnostics only as non-operational evidence of score-orientation instability

Note: IDS, intrusion detection system; PR-AUC, precision–recall area under the curve; ROC-AUC, receiver operating characteristic area under the curve; FPR, false-positive rate; ECE, expected calibration error; CNN, convolutional neural network; RNN, recurrent neural network; SSL, self-supervised learning.

1.2. Contributions

Accordingly, the study contributes an integrated evaluation methodology rather than a new detector architecture. Its specific contributions are as follows:

1. We construct two harmonized payload-free feature tiers: Rich-64 for detailed HIKARI-2021/CICIDS2017 analysis and Minimal-13 for three-way comparison with the CICIoT2023 sample subset.
2. We quantify the contrast between matched in-domain operation and strict source-only transfer, showing that usable in-domain 5% FPR points do not imply stable target-domain alerting.
3. We jointly assess source-derived calibration and threshold transfer with label-free target-score alert budgets, separating ranking from operational alert behavior.
4. We explicitly distinguish as-is operational scores from oracle-flipped diagnostics and identify orientation instability as a cross-dataset failure mode.

5. We provide qualified ablation evidence showing that masked-feature self-supervised pretraining does not consistently resolve benchmark-level transfer failure under the evaluated protocols.

2. Materials and Methods

2.1. Study Design and Evaluation Overview

This study is designed as a benchmark-shift-aware evaluation of network intrusion detection systems rather than as a single-model performance comparison. The evaluation examines whether IDS models, calibrated thresholds, and alerting behavior remain reliable when training and deployment benchmarks differ. The workflow consists of five stages: dataset harmonization, in-domain operating-point validation, strict cross-dataset transfer, alert-budget diagnostics, and score-orientation analysis.

Two evaluation tiers are used. The first tier uses a rich 64-feature payload-free schema shared by HIKARI-2021 and CICIDS2017. This tier supports detailed in-domain, chronological, and strict cross-dataset analysis. The second tier uses a minimal 13-feature schema shared by HIKARI-2021, CICIDS2017, and the CICIoT2023 sample subset. This tier supports three-way leave-one-dataset-out evaluation under a more restrictive common feature space.

Figure 1 summarizes the benchmark-shift-aware IDS evaluation framework, including harmonized feature construction, in-domain validation, strict cross-dataset source-only transfer, calibrated threshold transfer, target-side alert-budget diagnostics, and score-orientation analysis.

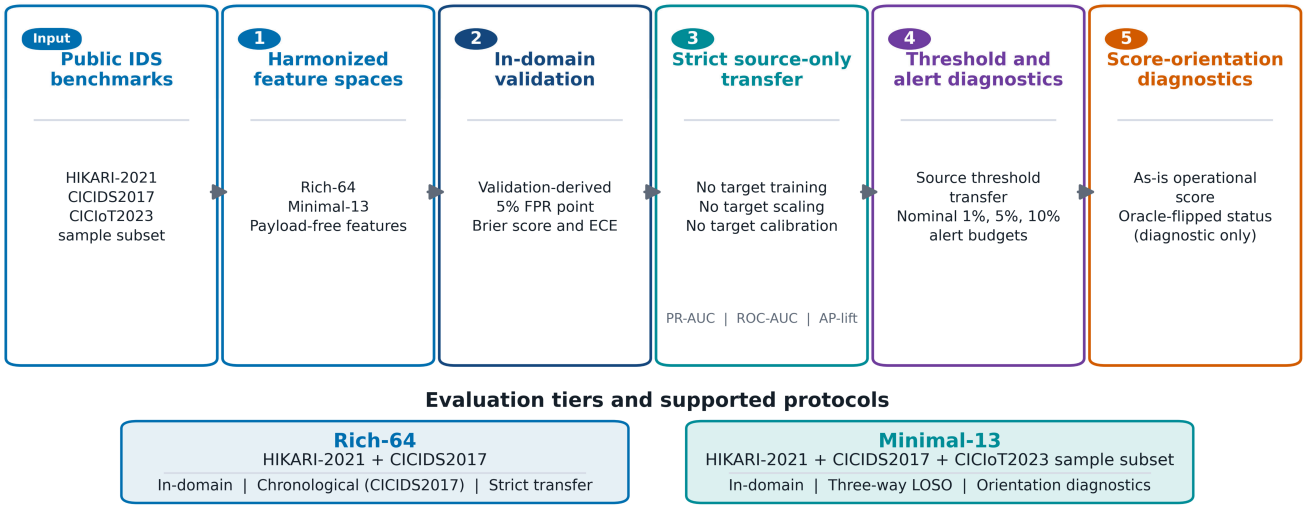


Figure 1. Overview of the benchmark-shift-aware IDS evaluation framework. The upper sequence shows the benchmark inputs followed by five stages: harmonized feature construction, in-domain validation, strict source-only transfer, threshold and alert diagnostics, and score-orientation diagnostics. The lower boxes identify benchmark coverage and support evaluation protocols for the Rich-64 and Minimal-13 feature spaces. Abbreviations: IDS, intrusion detection system; LOSO, leave-one-dataset-out.

2.2. Datasets and Harmonized Feature Spaces

Three public intrusion-detection benchmarks are used: HIKARI-2021, CICIDS2017, and the CICIoT2023 sample subset. These datasets differ in traffic-generation scenario, feature-extraction pipeline, label distribution, and attack prevalence. To reduce avoidable feature mismatch, the study constructs two harmonized payload-free feature spaces [6–8].

The Rich-64 schema is used for the main HIKARI-2021/CICIDS2017 analysis and contains 64 semantically aligned flow-level statistics. The Minimal-13 schema is used for three-way comparison and contains 13 features common to HIKARI-2021, CICIDS2017,

and the CICIoT2023 sample subset. In both schemas, labels are harmonized into a binary classification scheme in which benign traffic is assigned to the negative class, and all attack categories are assigned to the positive class. The full machine-readable Rich-64 and Minimal-13 feature lists are provided in Supplementary Table S1.

The CICIoT2023 sample subset used in the Minimal-13 tier was prepared from the processed project files Merged01.csv, Merged02.csv, and Merged03.csv, yielding 2,158,141 harmonized flow records. The subset contains 50,674 benign records and 2,107,467 attack records, corresponding to an attack prevalence of 97.6520%. No stochastic row-level downsampling is claimed for this sample-construction step; the subset is treated as a deterministic attack-heavy stress case, and conclusions involving CICIoT2023 are limited to sample-based benchmark-shift evidence rather than full-dataset generalization. Table 2 summarizes the datasets, sample counts, harmonized feature tiers, and evaluation roles used in this study.

Table 2. Dataset and harmonized feature-space summary.

Dataset	Samples	Benign Samples	Attack Samples	Attack Prevalence	Schema Tier	Main Role in This Study
HIKARI-2021	555,278	517,582	37,696	6.7887%	Rich-64; Minimal-13	In-domain; Rich-64 transfer; Minimal-13 LOSO
CICIDS2017	2,827,876	2,271,320	556,556	19.6811%	Rich-64; Minimal-13	In-domain; chronological; Rich-64 transfer; Minimal-13 LOSO
CICIoT2023 sample subset	2,158,141	50,674	2,107,467	97.6520%	Minimal-13	Three-way LOSO stress test under extreme prevalence imbalance
Rich-64 tier	HIKARI-2021; CICIDS2017	—	—	—	64 features	In-domain 5% FPR; CICIDS2017 chronological; HIKARI-2021 ↔ CICIDS2017 strict transfer
Minimal-13 tier	HIKARI-2021; CICIDS2017; CICIoT2023 sample subset	—	—	—	13 features	In-domain 5% FPR; three-way LOSO; score-orientation diagnostics

Note: LOSO, leave-one-dataset-out.

2.3. Preprocessing and Integrity Audit

All preprocessing is performed with training-only information whenever fitted statistics are required. Missing or invalid values are handled before modeling, skewed variables are transformed using log(1 + x), and continuous variables are standardized within the relevant source training split. Fields that can trivially leak identity and labels or capture metadata are excluded from the model input.

Because harmonization itself can introduce artifacts, the Rich-64 HIKARI-2021 artifact is subjected to a column-level integrity audit against the upstream stage2 log1p file. This audit identified several scale-inconsistent columns and produced a corrected HIKARI-2021 Rich-64 artifact for subsequent Rich-64 experiments. All Rich-64 results reported after this stage use the corrected HIKARI-2021 file together with the corresponding CICIDS2017 harmonized file.

2.4. In-Domain Operating-Point Evaluation

Before evaluating benchmark shifts, each harmonized schema is first validated under in-domain settings. The purpose of this stage is to establish that the feature spaces and baseline pipelines are not intrinsically weak within their own benchmarks. The primary operating point is selected under a 5% false-positive-rate budget using validation data, and the selected threshold is then applied to the test partition.

For each dataset and schema, the in-domain evaluation reports F1-score, precision, recall, observed false-positive rate, Brier score, and expected calibration error (ECE). Sigmoid calibration is used as the default calibrator, while isotonic calibration is retained as a sensitivity analysis. These in-domain operating points serve as the reference against which cross-dataset degradation is later interpreted.

2.5. Chronological and Cross-Dataset Protocols

For chronological drift evaluation, CICIDS2017 is split at the day level. Monday, Tuesday, and Wednesday are used for training, Thursday is used for validation, and Friday is used for testing. This protocol evaluates temporal robustness under a realistic shift in traffic and attack prevalence. HIKARI-2021 is not used for chronological drift evaluation because the current harmonized artifacts do not retain trustworthy time metadata; instead, HIKARI-2021 is used for stratified in-domain and cross-dataset evaluation. The chronological protocol follows the broader motivation of dataset-shift and concept-drift evaluation in non-stationary environments [14–17].

Strict cross-dataset evaluation is performed without using target-domain samples for training, scaling, calibration, or threshold selection. In the Rich-64 tier, the transfer directions are HIKARI-2021 to CICIDS2017 and CICIDS2017 to HIKARI-2021. In the Minimal-13 tier, a leave-one-dataset-out protocol is used across HIKARI-2021, CICIDS2017, and the CICIoT2023 sample subset. For each cross-dataset fold, thresholds and calibrators are fitted only on the source validation split and are transferred unchanged to the target test split.

The target prevalence values reported in the strict transfer tables are computed on the held-out target evaluation partition used for each transfer direction; these values can differ from the overall dataset-level prevalence reported in Table 2.

2.6. Calibration, Threshold Transfer, and Alert-Budget Diagnostics

Calibration is evaluated as part of the IDS operating procedure rather than as a purely post hoc visualization. Sigmoid calibration is fitted on the source validation split and applied to the corresponding test or target-domain scores. The primary operating rule selects a threshold on the validation negatives to target a 5% false-positive-rate budget. Within strict-transfer folds, the same source-validation partition provides the calibration scores and the negative-score quantile used for threshold selection; target data remain fully disjoint. This design is motivated by the calibration literature, which shows that probability estimates and ranking scores can behave differently under distribution shift [18–21]. Full calibration-quality comparisons are provided in Supplementary Table S2.

To distinguish threshold transfer failure from the absence of a ranking signal, the study also evaluates nominal target-side alert budgets of 1%, 5%, and 10%. For budget b, the operational cutoff is the empirical $(1 - b)$ quantile of target-domain scores; all records tied at the cutoff are included. Consequently, the realized alert fraction can slightly exceed the nominal budget. Target labels are not used to set the cutoff and are revealed only afterward to compute precision, recall, F1-score, and false-positive rate. This diagnostic approximates analyst-limited alert review and tests whether a residual ranking signal remains when source thresholds fail.

For target-side alert budgets, the false-positive rate is computed from the quantile-threshold-selected target alerts, including all ties at the cutoff, after evaluation labels are revealed. Let $\widetilde{b} = \frac{n_{\text{alerts}}}{n}$ denote the realized alert fraction after including all samples tied at the cutoff. If target attack prevalence is p and recall is r, then $FPR = \frac{\widetilde{b} - rp}{1 - p}$. This identity is used only for reporting diagnostic metrics, not for threshold selection.

2.7. Score-Orientation Diagnostic Protocol

Cross-dataset models may preserve some target-domain ranking signal while assigning higher scores to the wrong class. To identify this failure mode, the study reports score-orientation diagnostics. The operational result always uses the original score direction learned from the source domain. An oracle-flipped score is computed only as a diagnostic indicator to determine whether a target-domain ranking signal exists in the opposite direction. Because score flipping requires target labels, oracle-flipped results are not interpreted as deployable improvements.

2.8. Models, Training Details, and Self-Supervised Variant

The study compares tree-based and neural IDS baselines under the same harmonized feature spaces. XGBoost is used as a strong classical baseline because it performs well on tabular flow-level features and provides a practical reference for operating-point analysis. XGBoost models are trained on the source training split, selected on the corresponding validation split, and calibrated using only source validation scores.

The supervised Transformer is trained from scratch on the same harmonized tabular features. Each scalar feature is represented as a feature token by combining a value projection with a feature-identity embedding, and the resulting token sequence is processed by a Transformer encoder followed by a pooled binary classification head. The supervised objective is binary cross-entropy, and checkpoints are selected using source validation metrics before calibration and threshold selection.

The masked-feature self-supervised Transformer variant, denoted Transformer + self-supervised learning (SSL), uses the same encoder family but first masks a subset of feature tokens and trains a reconstruction head to predict the masked feature values. The pretrained encoder is then fine-tuned with the supervised binary objective. This variant is included to test whether reconstruction-oriented pretraining improves robustness under benchmark shift; it is not assumed to be superior a priori [11–13].

All reported model variants are treated as evaluation subjects within the benchmark-shift framework. To make the Rich-64 strict-transfer runs computationally tractable, supervised source-training sets were capped at 600,000 records using class-stratified sampling with seed 2026; smaller source sets were retained in full. SSL pretraining used at most 300,000 source records. The chronological CICIDS2017 XGBoost run used the same 600,000-record cap, whereas the chronological Transformer used the full 1,666,532-record training split. Experiment-specific model hyperparameters, optimization settings, training limits, random seeds, split summaries, calibration configuration, and software versions are reported in Supplementary Table S5. The reported values were obtained from fixed, single-seed experimental runs and are not presented as estimates from repeated runs.

2.9. Evaluation Metrics, Implementation, and Reproducibility

In this study, PR-AUC denotes non-interpolated average precision (AP) computed with scikit-learn's average_precision_score rather than trapezoidal integration of the precision–recall curve.

The primary threshold-free metrics are PR-AUC and ROC-AUC. Because attack prevalence differs substantially across datasets, AP-lift is also reported as PR-AUC minus the

target attack prevalence. Formally, for target labels y_t, source-trained target scores s_t, and target attack prevalence π_t, AP − lift = AP$(y_t, s_t) - \pi_t$.

The oracle orientation diagnostic is defined as $AUC_{oracle} = \max\{\text{ROC} - \text{AUC}(y_t, s_t), \text{ROC} - \text{AUC}(y_t, 1 - s_t)\}$, while all operational results retain s_t. Threshold-dependent metrics include F1-score, precision, recall, and false-positive rate under source-derived 5% FPR thresholds and target-side alert budgets. Calibration is summarized with the Brier score and ECE.

All reported results were generated from synchronized machine-readable result artifacts, including CSV and JSON summaries for operating points, cross-dataset comparisons, alert-budget diagnostics, orientation diagnostics, chronological evaluation, and Minimal-13 LOSO analysis. This artifact-based workflow was used to maintain consistency between result tables, discussion claims, and figure captions.

The computational environment used for the final result generation was Python 3.10.12 with NumPy 1.26.4, pandas 2.3.3, scikit-learn 1.7.2, XGBoost 3.2.0, PyTorch 2.7.1+cu118, CUDA 11.8, and an NVIDIA A100 80 GB GPU (NVIDIA Corporation, Santa Clara, CA, USA). The complete software environment and experiment-specific model configurations are documented in Supplementary Table S5.

3. Results

Results progress from matched in-domain operation to increasingly difficult benchmark shifts. Each subsection states the principal deployment-relevant observation, while the tables retain full numerical precision. Unless noted otherwise, values are from fixed single-seed runs, as specified in Section 2.8.

3.1. In-Domain Operating Points Under a 5% FPR Budget

Table 3 confirms near-target operation in all five calibrated XGBoost in-domain settings: observed FPRs ranged from 4.94% to 5.40%, while F1 ranged from 0.507 (Minimal-13 HIKARI-2021) to 0.995 (Minimal-13 CICIoT2023 sample subset). Thus, the harmonized feature spaces are usable within their own benchmarks; subsequent failures primarily reflect transfer rather than an intrinsically non-discriminative pipeline.

Table 3. In-domain 5% FPR operating points for the calibrated XGBoost baseline. Thresholds are selected on validation negatives and applied unchanged to the corresponding test split.

Schema	Dataset	Default Calibrator	Threshold	F1	Precision	Recall	Observed FPR	Brier	ECE
Rich-64	HIKARI-2021	sigmoid	0.313162	0.513668	0.458929	0.583234	0.050079	0.042475	0.003995
Rich-64	CICIDS2017	sigmoid	0.000393	0.907911	0.831371	0.999973	0.049700	0.000930	0.000248
Minimal-13	HIKARI-2021	sigmoid	0.293090	0.506906	0.456663	0.569572	0.049354	0.042631	0.003030
Minimal-13	CICIDS2017	sigmoid	0.067035	0.902187	0.827409	0.991825	0.050695	0.015660	0.003005
Minimal-13	CICIoT2023 sample subset	sigmoid	0.725514	0.994619	0.998692	0.990579	0.053971	0.006225	0.000397

3.2. Rich-64 Strict Cross-Dataset Transfer

Table 4 reports strict Rich-64 transfer with training, preprocessing, calibration, and threshold selection confined to the source benchmark.

Under the HIKARI-2021-to-CICIDS2017 transfer, all three models achieved zero recall on target attacks at their source-derived thresholds. The Transformer ranked best, but its PR-AUC of 0.411583 was almost identical to target prevalence (0.410954; AP-lift = 0.000629); XGBoost and Transformer + SSL had negative AP-lift and inverted score orientation.

Table 4. Strict Rich-64 cross-dataset transfer. Models are fitted and thresholded on the source benchmark only. AP-lift is PR-AUC minus target prevalence; all discrimination values use as-is operational scores, while 'flipped' is an oracle-only orientation diagnostic.

Transfer Direction	Model	Target Prevalence	PR-AUC	AP-Lift	ROC-AUC	Score Orientation	Recall at Source 5% FPR	Target FPR at Source 5% FPR
HIKARI-2021 → CICIDS2017	XGBoost	0.410954	0.300587	−0.110368	0.214126	flipped	0.000000	0.000000
HIKARI-2021 → CICIDS2017	Transformer	0.410954	0.411583	0.000629	0.588790	as-is	0.000000	0.000000
HIKARI-2021 → CICIDS2017	Transformer + SSL	0.410954	0.303289	−0.107666	0.311392	flipped	0.000000	0.000000
CICIDS2017 → HIKARI-2021	XGBoost	0.067885	0.094649	0.026764	0.613631	as-is	0.107839	0.189196
CICIDS2017 → HIKARI-2021	Transformer	0.067885	0.073486	0.005602	0.583881	as-is	0.000000	0.172619
CICIDS2017 → HIKARI-2021	Transformer + SSL	0.067885	0.080181	0.012296	0.627747	as-is	0.000133	0.083011

In the reverse direction, XGBoost retained the strongest ranking signal (PR-AUC = 0.094649; AP-lift = 0.026764), but its transferred threshold raised the target FPR to 18.9%. The neural variants provided negligible threshold-level recall. Figure 2 separates PR-AUC from prevalence and AP-lift from zero so that these directional failures are visible without repeating Table 4.

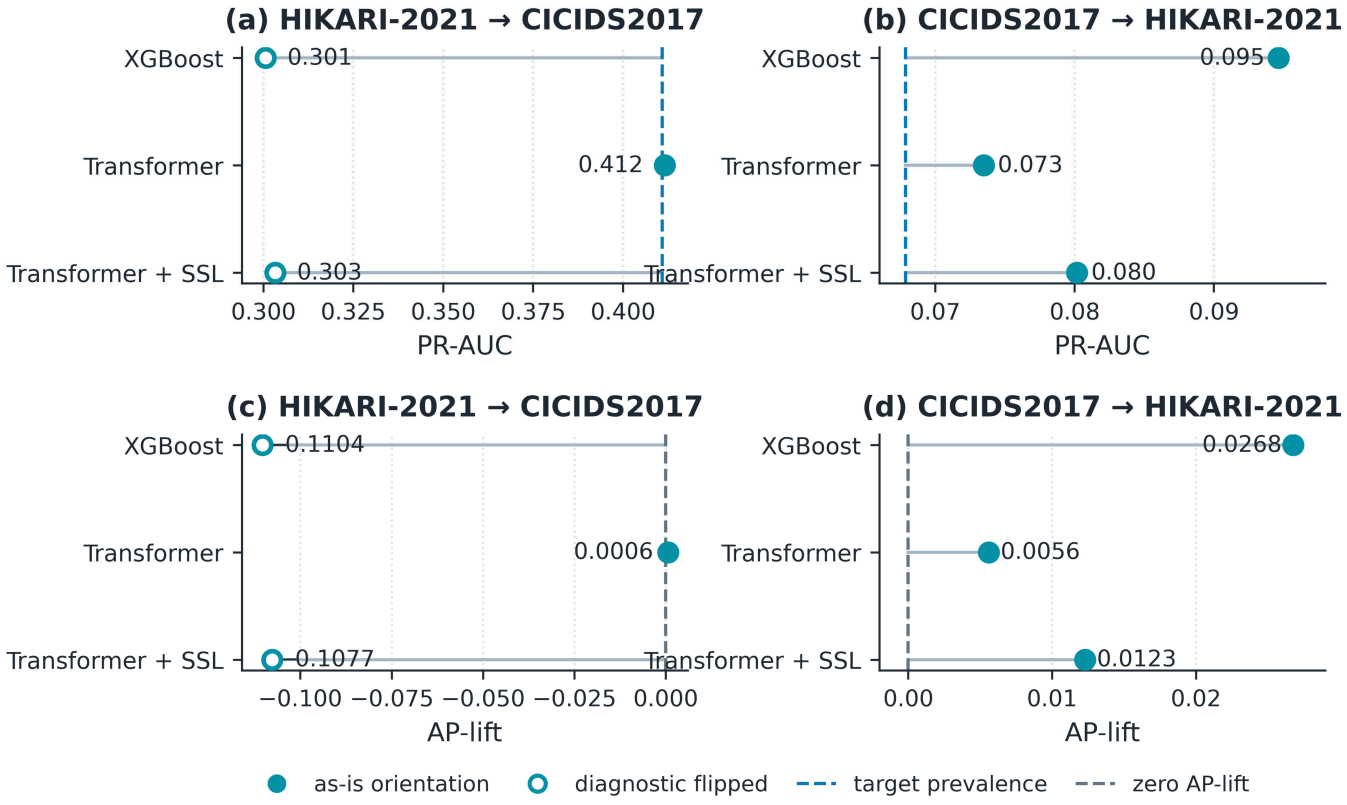


Figure 2. Strict Rich-64 cross-dataset discrimination. Panels (**a**,**b**) compare target PR-AUC with prevalence; panels (**c**,**d**) show AP-lift relative to zero. Filled and open markers denote as-is and diagnostic flipped orientation, respectively; all plotted values use as-is scores. Labels are rounded for readability; full-precision values and transferred-threshold behavior are reported in Table 4.

Thus, feature-level harmonization alone did not preserve score meaning or operating thresholds across benchmarks.

3.3. Target-Side Alert-Budget Diagnostics

Table 5 tests whether target-score quantile budgets reveal useful rankings after source-threshold failure. The cutoff uses target scores but no target labels; labels are revealed only for evaluation, and ties may make the realized alert fraction slightly exceed the nominal budget.

Table 5. Selected nominal target-side alert-budget diagnostics under strict cross-dataset transfer. For each source–target setting, the cutoff is the empirical $(1 - b)$ target-score quantile for nominal budget b; all records tied at the cutoff are alerted, so the realized alert fraction can slightly exceed b. Target labels are not used to select cutoffs and are used only post hoc to compute the reported metrics. Full nominal 1%, 5%, and 10% results are provided in Supplementary Table S3.

Transfer Direction	Model	Alert Budget	Precision	Recall	F1	FPR
CICIDS2017 → HIKARI-2021	XGBoost	nominal 1%	0.259226	0.038201	0.066590	0.007950
CICIDS2017 → HIKARI-2021	XGBoost	nominal 5%	0.129418	0.099085	0.112238	0.048543
CICIDS2017 → HIKARI-2021	XGBoost	nominal 10%	0.069378	0.103860	0.083187	0.101462
HIKARI-2021 → CICIDS2017	XGBoost	nominal 5%	0.012409	0.001510	0.002692	0.083830
HIKARI-2021 → CICIDS2017	Transformer	nominal 10%	0.133709	0.032536	0.052337	0.147067
HIKARI-2021 → CICIDS2017	Transformer + SSL	nominal 10%	0.000484	0.000118	0.000189	0.169684

Only the CICIDS2017-to-HIKARI-2021 XGBoost transfer showed meaningful residual utility: at the nominal 5% budget, precision was 0.129, recall 0.099, F1 0.112, and FPR 4.85%. Performance was weak for the other models in the CICIDS2017-to-HIKARI-2021 direction and for all models in the reverse direction (Figure 3).

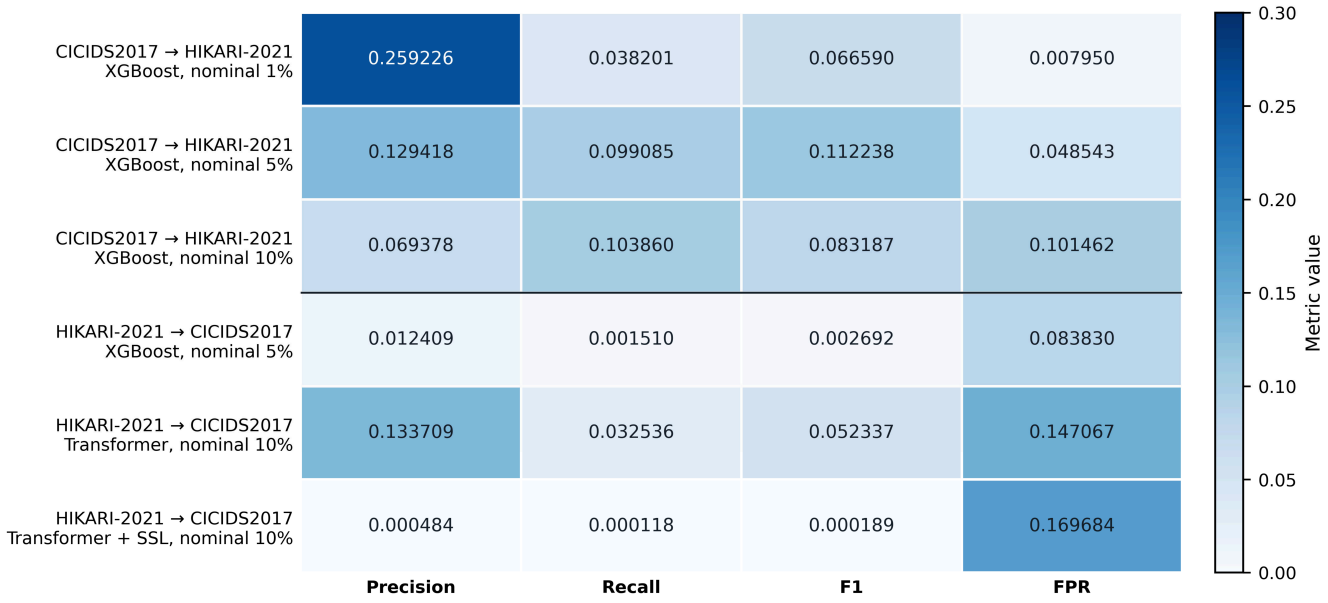


Figure 3. Selected nominal target-side alert-budget diagnostics under strict cross-dataset transfer. Nominal budgets are implemented with empirical target-score quantiles and include all ties at the cutoff; target labels are used only post hoc for metric computation. The heatmap highlights residual ranking signal in the CICIDS2017-to-HIKARI-2021 XGBoost transfer and weak operational behavior in the reverse direction.

Alert budgeting can limit review volume and diagnose residual ranking, but it does not repair cross-benchmark failure.

3.4. Minimal-13 Three-Way LOSO Transfer

The Minimal-13 LOSO test broadens transfer to all three datasets under the most restrictive common feature set (Table 6).

Table 6. Minimal-13 three-way LOSO contrast between matched in-domain recall and source-only transfer. Calibration and thresholds are fitted on source validation data only; discrimination values use as-is scores, while 'flipped' is an oracle-only diagnostic.

Target Dataset	Status	Target Prevalence	In-Domain Recall at 5% FPR	LOSO PR-AUC	AP-Lift	ROC-AUC	Score Orientation	Recall at Source 5% FPR
CICIDS2017	orientation inverted	0.196811	0.991825	0.150204	−0.046607	0.366245	flipped	0.000000
CICIoT2023 sample subset	orientation inverted	0.976520	0.990579	0.924835	−0.051685	0.046086	flipped	0.000000
HIKARI-2021	near chance	0.067887	0.569572	0.067786	−0.000101	0.537855	as-is	0.031595

No held-out target exhibited all three of the following: positive AP-lift, stable orientation, and useful source-threshold recall. CICIDS2017 and CICIoT2023 both had negative AP-lift (−0.046607 and −0.051685), inverted orientation, and zero recall. HIKARI-2021 was near chance (AP-lift = −0.000101) with recall = 0.031595.

Figure 4 makes the pattern explicit: in-domain recall is high, yet LOSO AP-lift is non-positive and transferred-threshold recall collapses for every target.

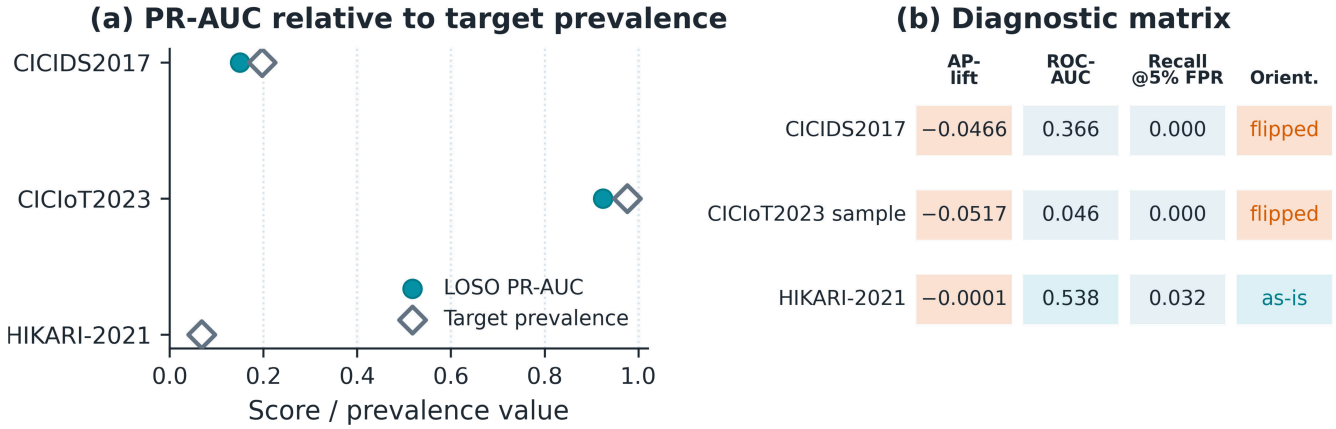


Figure 4. Minimal-13 three-way LOSO diagnostics. Panel (**a**) compares as-is LOSO PR-AUC with target prevalence. Panel (**b**) reports AP-lift, ROC-AUC, source-threshold recall, and diagnostic orientation. Labels are rounded for readability; full-precision values are reported in Table 6. Flipped orientation is an oracle-only diagnostic.

3.5. Chronological Benchmark Shift on CICIDS2017

Day-level evaluation uses Monday–Wednesday training, Thursday validation, and Friday testing, so ranking and transferred operating behavior are assessed under temporal shift.

Relative to XGBoost, the supervised Transformer improved Friday ranking (PR-AUC: 0.790 versus 0.669; ROC-AUC: 0.891 versus 0.820; Table 7).

At the Thursday-derived 5% FPR threshold, however, XGBoost produced F1 = 0.494, recall = 0.362, and Friday FPR = 7.27%; the Transformer kept FPR at 4.68% but produced F1 = 0.173 and recall = 0.101. Figure 5, therefore, separates improved ranking from weaker threshold-level sensitivity.

Table 7. CICIDS2017 chronological shift: Friday ranking and operating behavior after transferring the Thursday-derived 5% FPR threshold.

Model	PR-AUC	ROC-AUC	Test-Day F1 at Validation-Derived Threshold	Test-Day Recall at Validation-Derived Threshold	Test-Day FPR
XGBoost	0.668610	0.819926	0.494044	0.362228	0.072662
Supervised Transformer	0.790004	0.891402	0.172801	0.100912	0.046776

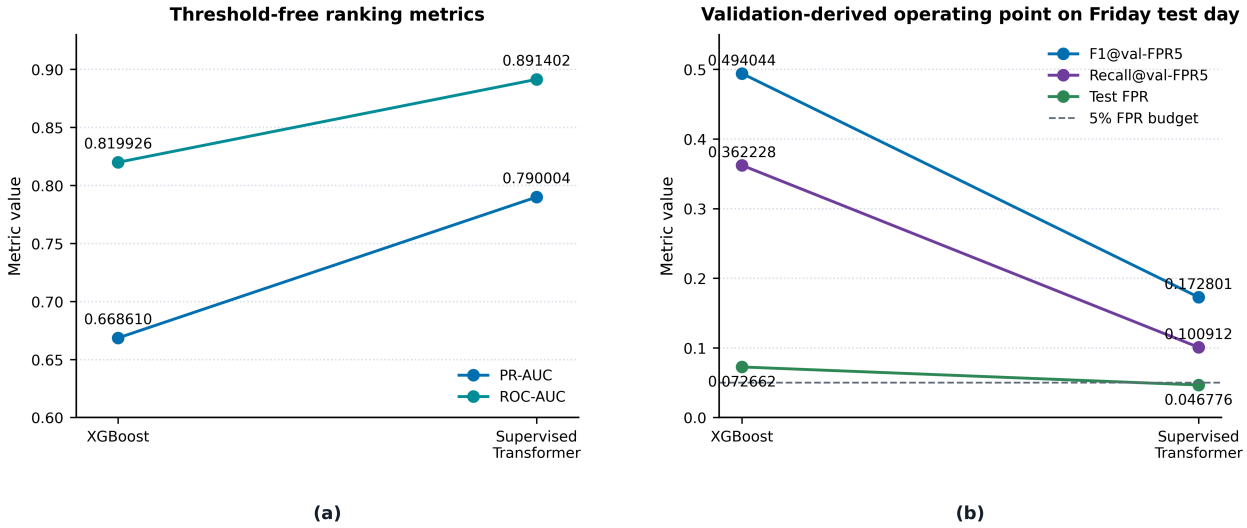


Figure 5. Chronological ranking and operating-point mismatch on CICIDS2017. Panel (**a**) compares threshold-free PR-AUC and ROC-AUC for XGBoost and the supervised Transformer on the Friday test day. Panel (**b**) compares F1-score, recall, and test-day FPR after transferring the validation-derived 5% FPR threshold; the dashed reference line marks the intended 5% FPR budget. Filled markers and connecting guides emphasize that improved ranking metrics do not necessarily imply stable FPR-budgeted operating behavior.

3.6. Score-Orientation Diagnostics

Orientation diagnostics exposed a qualitatively different failure: some source-trained scores ranked the wrong target class higher. The clearest case was HIKARI-2021-to-CICIDS2017 XGBoost, whose as-is alert-budget performance was poor although oracle inversion revealed target-domain structure. Because inversion requires target labels, it is not an operational remedy. Figure 6 summarizes orientation status across Rich-64 and Minimal-13 transfers; full values are in Supplementary Table S4.

3.7. Summary of Empirical Findings

Three conclusions recur across the protocols. First, matched operation near a 5% FPR does not predict strict-transfer usability. Second, a benchmark shift can alter ranking, calibration, threshold behavior, and even score orientation. Third, neither target-score budgets nor the evaluated deep and self-supervised variants consistently restore operational performance. Robustness claims should therefore report all five diagnostic levels rather than a single AUC.

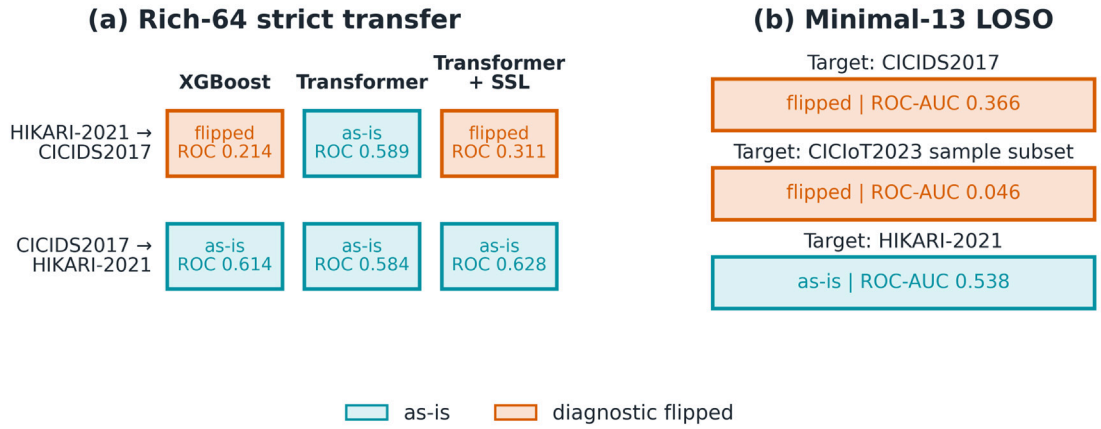


Figure 6. Score-orientation instability under benchmark shift. Panel (**a**) summarizes Rich-64 strict-transfer settings by model; panel (**b**) summarizes Minimal-13 LOSO targets. Cells show the diagnostic orientation and as-is ROC-AUC rounded to three decimals; full-precision values are reported in Tables 4 and 6 and Supplementary Table S4. Flipped orientation is not deployable without target labels.

4. Discussion

4.1. Benchmark Shift as an Operational IDS Problem

Matched in-domain tests established that the harmonized schemas and baseline pipeline can support 5% FPR operation. Under strict transfer, however, HIKARI-2021-to-CICIDS2017 thresholds yielded zero recall, while reverse-direction XGBoost inflated the target FPR to 18.9%. Benchmark shift, therefore, affected not only discrimination but also the operational mapping from scores to alerts.

These failures distinguish benchmark robustness from single-dataset performance. A complete evaluation must pair threshold-free metrics with calibration, transferred-threshold behavior, analyst-budgeted alerting, and orientation checks; otherwise, near-prevalence ranking, threshold collapse, or score inversion can remain hidden.

4.2. Calibration Is Necessary but Not Sufficient

Sigmoid calibration supported stable in-domain points, but source-fitted probability mappings did not remain reliable under altered prevalence and traffic composition. A source threshold can become too conservative, suppressing attacks, or too permissive, producing excess false alarms. Calibration should therefore be assessed by Brier score and ECE together with transferred thresholds and target-side budgets, not by in-domain calibration metrics alone.

4.3. Alert-Budget Diagnostics Reveal Residual but Limited Signal

Target-side budgets answer a different question from source-threshold transfer: whether the top-ranked target events contain a useful concentration of attacks under a fixed review capacity when the numeric source cutoff is unusable.

Only CICIDS2017-to-HIKARI-2021 XGBoost retained meaningful budgeted utility; most reverse-direction settings did not. Budgets control workload without certifying recall, FPR, or calibration; they serve as a diagnostic, not as an adaptation mechanism. This interpretation aligns with recent work on cross-dataset and temporal generalization, target-FPR-constrained evaluation, and false-positive-aware prioritization [22–24].

4.4. Score-Orientation Instability as an Underreported Failure Mode

Score inversion is distinct from ordinary performance loss: a detector may preserve target-domain structure while assigning higher scores to the wrong class. Oracle inversion

can expose this structure but cannot be deployed because it requires target labels. Reporting the as-is result together with diagnostic orientation prevents both hidden failure and overstated performance.

4.5. Qualified Findings on Transformer and Self-Supervised Variants

Chronological results show that neural models can improve ranking without improving the selected deployment point: the Transformer increased PR-AUC and ROC-AUC but transferred conservatively with low recall, whereas XGBoost exceeded the FPR budget. Both threshold-free and threshold-dependent behavior are therefore needed to assess temporal robustness.

Likewise, masked-feature SSL did not consistently stabilize strict transfer. Because SSL was not evaluated chronologically, no temporal robustness claim was made. These are qualified ablation findings, not evidence against stronger SSL objectives.

4.6. Implications for IDS Research and Deployment

The framework implies three minimum practices for IDS research: report matched and shifted tests side by side, audit both source-threshold transfer and label-free score-quantile budgets, and test orientation when discrimination and operating metrics diverge. It complements recent deployment-oriented and cross-protocol evidence by integrating these checks in a common harmonized protocol [22–25].

Security operations centers can apply the framework as a staged pre-deployment audit. First, establish source-domain discrimination, calibration, and a 5% FPR operating point. Next, run the detector in shadow mode on the new network and compare score distributions and alert volume with the source baseline. If the source threshold becomes unstable, a target-score quantile budget can cap analyst workload without using labels to set the cutoff; however, it does not certify recall or FPR. Once analyst adjudications or delayed labels are available, re-estimate recall, FPR, Brier score, and ECE, and check score orientation. Persistent degradation should trigger recalibration or retraining. Oracle flipping remains diagnostic only, while continuous drift monitoring should determine when to repeat the audit.

4.7. Limitations and Threats to Validity

This study has several limitations. First, CICIoT2023 is represented by a deterministic sample subset, so the Minimal-13 findings provide only sample-based evidence under extreme prevalence imbalance. Second, chronology is evaluated only on CICIDS2017 because trustworthy time metadata were unavailable in the current HIKARI-2021 artifacts. Third, the evaluated self-supervised objective is intentionally lightweight; stronger objectives may behave differently.

Fourth, oracle-flipped results identify inversion but are not operational. Fifth, comparisons use fixed single-seed runs, so repeated-seed uncertainty is unquantified. Sixth, strict-transfer calibration and threshold selection share one source-validation partition; a separate calibration split would be stricter. Harmonization may also remove dataset-specific signals useful to specialized detectors. These constraints delimit the framework's evidence and preclude claims of universal model superiority.

Computationally, this is an offline diagnostic framework rather than a new real-time detector. Repeating Transformer and SSL training across source–target combinations requires more computation and storage than the XGBoost baseline; tractability was supported by training caps and an NVIDIA A100 80 GB GPU. Wall-clock time, peak memory, energy use, and online throughput were not benchmarked, so efficiency on lower-resource systems remains unknown. Applying the framework elsewhere also requires semantically alignable payload-free features, consistent binary labels, and reliable timestamps where chrono-

logical testing is intended. Production use additionally requires stable feature pipelines, delayed-label handling, and ongoing drift monitoring.

5. Conclusions

This study evaluated benchmark-shift-aware intrusion detection through matched, chronological, and strict cross-dataset protocols using Rich-64 and Minimal-13 payload-free schemas. This study contributes an integrated diagnostic methodology for evaluating discrimination, calibration, source-threshold transfer, label-free alert budgets, and score orientation; it does not propose a new detector architecture.

In-domain XGBoost operated near the intended 5% FPR in all five settings (4.94–5.40%), with F1 ranging from 0.507 to 0.995. Under strict HIKARI-2021-to-CICIDS2017 transfer, every model had zero recall at its source threshold, and two models showed inverted orientation. In the reverse direction, XGBoost reached an 18.9% target FPR, while a nominal 5% target-side budget yielded F1 = 0.112. Chronologically, the Transformer improved ranking but produced lower threshold-level recall. These results show that benchmark shifts can disrupt rankings, calibrated scores, thresholds, and workload simultaneously.

Future work should test class-conditional alignment designed to preserve source label semantics, including domain-adversarial objectives [25] and correlation-alignment losses [26], together with shift-aware importance weighting and periodically obtained analyst-validated labels for lightweight recalibration and threshold re-estimation. Drift detectors could trigger these updates when score distributions or alert volumes change. Candidate methods should be evaluated under the same strict-transfer, chronological, calibration, workload, and orientation protocols, with negative transfer reported alongside gains. Extensions should also use full-scale IoT benchmarks, additional traffic-generation regimes, repeated seeds, and explicit computational-efficiency measurements.

Supplementary Materials: The following supporting information can be downloaded at https://www.mdpi.com/article/10.3390/electronics15173761/s1. Table S1: Complete Rich-64 and Minimal-13 feature lists; Table S2: Calibration-quality comparison including sigmoid and isotonic sensitivity results; Table S3: Full nominal target-side alert-budget diagnostics; Table S4: Score-orientation diagnostics; Table S5: Experiment-specific model hyperparameters, optimization settings, training limits, random seeds, split summaries, calibration configuration, and software environment. Machine-readable CSV versions of these tables are provided in Supplementary Materials.

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Data Availability Statement: The public datasets analyzed in this study are available from their respective public repositories: CICIDS2017 from the Canadian Institute for Cybersecurity (https://www.unb.ca/cic/datasets/ids-2017.html; accessed on 2 April 2026), HIKARI-2021 from its Zenodo dataset release (https://zenodo.org/records/5199540; accessed on 2 April 2026), and CICIoT2023 from the Canadian Institute for Cybersecurity (https://www.unb.ca/cic/datasets/iotdataset-2023.html; accessed on 2 April 2026). Machine-readable supplementary tables supporting the reported results are provided in Supplementary Materials. Raw public datasets are not redistributed. Further inquiries can be directed to the corresponding author.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Pérez-Jove, R.; Munteanu, C.R.; Dorado, J.; Pazos, A.; Vázquez-Naya, J. Network Traffic Foundation Models: A Systematic Review. *Comput. Netw.* **2026**, *276*, 111998. [CrossRef]
2. Sattar, S.; Khan, S.; Khan, M.I.; Akhmediyarova, A.; Mamyrbayev, O.; Kassymova, D.; Oralbekova, D.; Alimkulova, J. Anomaly Detection in Encrypted Network Traffic Using Self-Supervised Learning. *Sci. Rep.* **2025**, *15*, 26585. [CrossRef] [PubMed]
3. Cai, S.; Tang, H.; Chen, J.; Hu, Y.; Guo, W. CDDA-MD: An Efficient Malicious Traffic Detection Method Based on Concept Drift Detection and Adaptation Technique. *Comput. Secur.* **2025**, *148*, 104121. [CrossRef]
4. de Nascimento, C.M.; Hou, J. Uncertainty-Aware Adaptive Intrusion Detection Using Hybrid CNN-LSTM with cWGAN-GP Augmentation and Human-in-the-Loop Feedback. *Safety* **2025**, *11*, 120. [CrossRef]
5. Tariq, S.; Baruwal Chhetri, M.; Nepal, S.; Paris, C. Alert Fatigue in Security Operations Centres: Research Challenges and Opportunities. *ACM Comput. Surv.* **2025**, *57*, 1–38. [CrossRef]
6. Sharafaldin, I.; Habibi Lashkari, A.; Ghorbani, A.A. Toward Generating a New Intrusion Detection Dataset and Intrusion Traffic Characterization. In Proceedings of the 4th International Conference on Information Systems Security and Privacy (ICISSP), Funchal, Portugal, 22–24 January 2018; pp. 108–116. [CrossRef]
7. Ferriyan, A.; Thamrin, A.H.; Takeda, K.; Murai, J. Generating Network Intrusion Detection Dataset Based on Real and Encrypted Synthetic Attack Traffic. *Appl. Sci.* **2021**, *11*, 7868. [CrossRef]
8. Neto, E.C.P.; Dadkhah, S.; Ferreira, R.; Zohourian, A.; Lu, R.; Ghorbani, A.A. CICIoT2023: A Real-Time Dataset and Benchmark for Large-Scale Attacks in IoT Environment. *Sensors* **2023**, *23*, 5941. [CrossRef] [PubMed]
9. Verkerken, M.; D'hooge, L.; Wauters, T.; Volckaert, B.; De Turck, F. Towards Model Generalization for Intrusion Detection: Unsupervised Machine Learning Techniques. *J. Netw. Syst. Manag.* **2022**, *30*, 12. [CrossRef]
10. Pinto, D.; Amorim, I.; Maia, E.; Praça, I. A Review on Intrusion Detection Datasets: Tools, Processes, and Features. *Comput. Netw.* **2025**, *262*, 111177. [CrossRef]
11. Wang, Y.; Xu, H.; Ma, J.; Jin, Q. MTRC: A Self-Supervised Network Intrusion Detection Framework Based on Multiple Transformers Enabled Data Reconstruction with Contrastive Learning. *J. Netw. Comput. Appl.* **2025**, *243*, 104300. [CrossRef]
12. Wang, W.; Jian, S.; Tan, Y.; Wu, Q.; Huang, C. Robust Unsupervised Network Intrusion Detection with Self-Supervised Masked Context Reconstruction. *Comput. Secur.* **2023**, *128*, 103131. [CrossRef]
13. Koukoulis, I.; Syrigos, I.; Korakis, T. Self-Supervised Transformer-Based Contrastive Learning for Intrusion Detection Systems. *arXiv* **2025**, arXiv:2505.08816. [CrossRef]
14. Gama, J.; Žliobaitė, I.; Bifet, A.; Pechenizkiy, M.; Bouchachia, A. A Survey on Concept Drift Adaptation. *ACM Comput. Surv.* **2014**, *46*, 1–37. [CrossRef] [PubMed]
15. Moreno-Torres, J.G.; Raeder, T.; Alaiz-Rodríguez, R.; Chawla, N.V.; Herrera, F. A Unifying View on Dataset Shift in Classification. *Pattern Recognit.* **2012**, *45*, 521–530. [CrossRef]
16. Quiñonero-Candela, J.; Sugiyama, M.; Schwaighofer, A.; Lawrence, N.D. (Eds.) *Dataset Shift in Machine Learning*; The MIT Press: Cambridge, MA, USA, 2008.
17. Sugiyama, M.; Kawanabe, M. *Machine Learning in Non-Stationary Environments: Introduction to Covariate Shift Adaptation*; The MIT Press: Cambridge, MA, USA, 2012. [CrossRef]
18. Guo, C.; Pleiss, G.; Sun, Y.; Weinberger, K.Q. On Calibration of Modern Neural Networks. In *Proceedings of the 34th International Conference on Machine Learning, Sydney, Australia, 6–11 August 2017*; Proceedings of Machine Learning Research: Cambridge, MA, USA, 2017; Volume 70, pp. 1321–1330.
19. Niculescu-Mizil, A.; Caruana, R. Predicting Good Probabilities with Supervised Learning. In Proceedings of the 22nd International Conference on Machine Learning (ICML), Bonn, Germany, 7–11 August 2005; pp. 625–632. [CrossRef]
20. Ovadia, Y.; Fertig, E.; Ren, J.; Nado, Z.; Sculley, D.; Nowozin, S.; Dillon, J.; Lakshminarayanan, B.; Snoek, J. Can You Trust Your Model's Uncertainty? Evaluating Predictive Uncertainty under Dataset Shift. In *Advances in Neural Information Processing Systems 32*; Curran Associates, Inc.: Red Hook, NY, USA, 2019; pp. 13991–14002.
21. Platt, J.C. Probabilities for SV Machines. In *Advances in Large-Margin Classifiers*; Smola, A.J., Bartlett, P.L., Schölkopf, B., Schuurmans, D., Eds.; The MIT Press: Cambridge, MA, USA, 2000; pp. 61–74. [CrossRef]
22. Elangovan, R.; Parthasarathy, D.D.; Jawahar, M.; Kaliyaperumal, P.; Balusamy, B.; Yogarayan, S.; Venkatesan, V. Cross-Dataset Temporal and Semantic Generalization of Intrusion Detection Models for the Future Internet. *Future Internet* **2026**, *18*, 194. [CrossRef]
23. Ha, Y.; Kim, K. Operationally Constrained Zero-Day Intrusion Detection with Target-FPR Calibration and Similarity Graph Construction. *Appl. Sci.* **2026**, *16*, 2284. [CrossRef]
24. Gombar, M. From Detection to Triage: Explainable Suspicious Flow Prioritization for Multiclass Intrusion Detection Using CSE-CIC-IDS2018. *Electronics* **2026**, *15*, 2739. [CrossRef]

25. Jin, H. Cross-Protocol Domain Gap in Internet of Things Intrusion and Anomaly Detection: An Empirical Internet Protocol-to-Bluetooth Low Energy Study of Domain-Adversarial Training. *Sensors* **2026**, *26*, 1184. [CrossRef] [PubMed]
26. Sun, B.; Saenko, K. Deep CORAL: Correlation Alignment for Deep Domain Adaptation. In *Computer Vision-ECCV 2016 Workshops*; Hua, G., Jegou, H., Eds.; Lecture Notes in Computer Science; Springer: Cham, Switzerland, 2016; Volume 9915, pp. 443–450. [CrossRef]

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